

M21_20260502

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Load libraries

Load Files

```
# Load data: rt words

rt_w <- read_csv("rt_words_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    Item   = factor(target),
    logRT  = log(response_time),
    OS     = as.numeric(Dim.2),
    LP     = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: rt nonwords

rt_nw <- read_csv("rt_nwords_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    ItemID = factor(ItemID),
    complexity = factor(complexity,
                        levels = c("SIMP", "COMP"),
                        labels = c("Simple", "Complex")),
    logRT  = log(response_time),
    OS     = as.numeric(Dim.2),
    LP     = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: erp_fs words

erp_fs_w <- read_csv("erp_fs_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
                        levels = c("Small", "Large")),
```

```

    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_fs nonwords
erp_fs_nw <- read_csv("erp_fs_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
                        levels = c("Small", "Large")),
    complexity = factor(complexity,
                        labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf words
erp_bf_w <- read_csv("erp_bf_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
                           levels = c("Low", "High")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf nonwords
erp_bf_nw <- read_csv("erp_bf_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
                           levels = c("Low", "High")),
    complexity = factor(complexity,
                        labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

```

WORDS

RESPONSE TIME DATA

```

library(lmerTest)

m_rt_words <- lmer(

```

```

logRT ~ zLogBF + zLogFS + OS + LP +
  (1 | SubjID) +
  (1 | Item),
data = rt_w,
control = lmerControl(optimizer = "bobyqa")
)
summary(m_rt_words)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: logRT ~ zLogBF + zLogFS + OS + LP + (1 | SubjID) + (1 | Item)
## Data: rt_w
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: -3536.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -9.8884 -0.6626 -0.0643  0.5951  5.1184
##
## Random effects:
## Groups Name Variance Std.Dev.
## Item (Intercept) 0.001807 0.04251
## SubjID (Intercept) 0.017718 0.13311
## Residual 0.029621 0.17211
## Number of obs: 5847, groups: Item, 100; SubjID, 67
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  6.390069   0.016965  72.434643 376.656 < 2e-16 ***
## zLogBF      -0.017184   0.004696 100.974129  -3.659 0.000404 ***
## zLogFS      -0.012858   0.004865  93.591065  -2.643 0.009629 **
## OS          -0.034712   0.018734  63.842905  -1.853 0.068517 .
## LP           0.011591   0.012386  63.973575   0.936 0.352893
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) zLogBF zLogFS OS
## zLogBF  0.009
## zLogFS  0.002 -0.105
## OS      0.012 -0.001  0.001
## LP      0.008  0.000  0.001  0.008

# Effect sizes for individual terms
eta_squared(m_rt_words, partial = TRUE)

## # Effect Size for ANOVA (Type III)
##
## Parameter | Eta2 (partial) | 95% CI
## -----
## zLogBF | 0.12 | [0.04, 1.00]
## zLogFS | 0.07 | [0.01, 1.00]
## OS | 0.05 | [0.00, 1.00]
## LP | 0.01 | [0.00, 1.00]

```

```
##
## - One-sided CIs: upper bound fixed at [1.00].
```

```
# Model R2
r2_nakagawa(m_rt_words)
```

```
## # R2 for Mixed Models
##
## Conditional R2: 0.417
## Marginal R2: 0.033
```

ERP DATA

N250 DATA

```
contrasts(erp_bf_w$base_frequency) <- c(-0.5, 0.5)

m_w_N250_bf <- lmer(
  value ~ base_frequency * OS + LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250_bf)
```

Base Frequency

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ base_frequency * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
## Data: subset(erp_bf_w, time_window == "N250")
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 18999.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.4496 -0.3820 -0.0038  0.3905 14.0821
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## SubjID:chlabel (Intercept) 19.723    4.441
## SubjID          (Intercept) 12.044    3.470
## Residual                        8.095    2.845
## Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -1.5059    0.4650   56.9999  -3.239  0.00201 **
## base_frequency1 -0.3183    0.1001 1618.0001  -3.181  0.00150 **
## OS              0.5341    0.5200   56.9999   1.027  0.30874
## LP             -0.5356    0.3781   56.9999  -1.417  0.16203
## base_frequency1:OS -0.4549    0.1120 1618.0001  -4.062 5.09e-05 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) bs_fr1 OS      LP
## bas_frncyl 0.000
## OS          -0.046  0.000
## LP          0.038  0.000  0.021
## bs_frqn1:OS 0.000 -0.047  0.000  0.000
# Effect sizes for individual terms
eta_squared(m_w_N250_bf, partial = TRUE)
```

```
## # Effect Size for ANOVA (Type III)
##
## Parameter          | Eta2 (partial) |      95% CI
## -----
## base_frequency    |      6.21e-03 | [0.00, 1.00]
## OS                 |      0.02 | [0.00, 1.00]
## LP                 |      0.03 | [0.00, 1.00]
## base_frequency:OS |      0.01 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

```
# Model R²
r2_nakagawa(m_w_N250_bf)
```

```
## # R2 for Mixed Models
##
##      Conditional R2: 0.801
##      Marginal R2: 0.018
```

```
contrasts(erp_fs_w$family_size) <- c(-0.5, 0.5)

m_w_N250_fs <- lmer(
  value ~ family_size * OS + LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250_fs)
```

Family Size

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ family_size * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
## Data: subset(erp_fs_w, time_window == "N250")
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 15026.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -8.4669 -0.4045 -0.0218  0.3824 12.9276
```

```
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
##   SubjID:chlabel (Intercept) 4.625    2.151
##   SubjID        (Intercept) 2.918    1.708
##   Residual              2.754    1.659
## Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.75979    0.22921   56.99995  -3.315   0.0016 **
## family_size1    0.32357    0.05837 1617.99993   5.543 3.46e-08 ***
## OS              0.26085    0.25635   56.99995   1.018   0.3132
## LP             -0.27860    0.18636   56.99995  -1.495   0.1404
## family_size1:OS  0.11120    0.06532 1617.99992   1.702   0.0889 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) fmly_1 OS      LP
## family_siz1  0.000
## OS           -0.046  0.000
## LP           0.038  0.000  0.021
## fmly_sz1:OS  0.000 -0.047  0.000  0.000

# Effect sizes for individual terms
eta_squared(m_w_N250_fs, partial = TRUE)

## # Effect Size for ANOVA (Type III)
##
## Parameter      | Eta2 (partial) |      95% CI
## -----
## family_size    |      0.02 | [0.01, 1.00]
## OS              |      0.02 | [0.00, 1.00]
## LP              |      0.04 | [0.00, 1.00]
## family_size:OS | 1.79e-03 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].

# Model R²
r2_nakagawa(m_w_N250_fs)

## # R2 for Mixed Models
##
##   Conditional R2: 0.738
##   Marginal R2: 0.019
```

N400 DATA

```
contrasts(erp_bf_w$base_frequency) <- c(-0.5, 0.5)

m_w_N400_bf <- lmer(
  value ~ base_frequency * LP + OS +
```

```

    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_bf)

```

Base Frequency

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ base_frequency * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
## Data: subset(erp_bf_w, time_window == "N400")
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 19468.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -11.6094  -0.3522   0.0107   0.3807  10.5439
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## SubjID:chlabel (Intercept) 23.249   4.822
## SubjID         (Intercept) 18.541   4.306
## Residual                        9.157   3.026
## Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    5.617e-01  5.722e-01 5.700e+01  0.982   0.3304
## base_frequency1 2.141e-01  1.064e-01 1.618e+03  2.012   0.0444 *
## LP              4.677e-02  4.652e-01 5.700e+01  0.101   0.9203
## OS              1.290e+00  6.399e-01 5.700e+01  2.016   0.0485 *
## base_frequency1:LP 3.444e-01  8.659e-02 1.618e+03  3.977 7.29e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) bs_fr1 LP      OS
## bas_frncyl1  0.000
## LP            0.038  0.000
## OS            -0.046  0.000  0.021
## bs_frqn1:LP  0.000  0.039  0.000  0.000

```

Effect sizes for individual terms

```

eta_squared(m_w_N400_bf, partial = TRUE)

```

```

## # Effect Size for ANOVA (Type III)
##
## Parameter          | Eta2 (partial) |          95% CI
## -----
## base_frequency    |          2.50e-03 | [0.00, 1.00]
## LP                 |          1.77e-04 | [0.00, 1.00]
## OS                 |           0.07 | [0.00, 1.00]

```

```
## base_frequency:LP |          9.68e-03 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

```
# Model R2
r2_nakagawa(m_w_N400_bf)
```

```
## # R2 for Mixed Models
##
## Conditional R2: 0.825
## Marginal R2: 0.026
```

```
contrasts(erp_fs_w$family_size) <- c(-0.5, 0.5)

m_w_N400_fs <- lmer(
  value ~ family_size * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_fs)
```

Family Size

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ family_size * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
## Data: subset(erp_fs_w, time_window == "N400")
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 15561.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -10.9307  -0.3874   0.0081   0.4071   6.2048
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## SubjID:chlabel (Intercept)  5.238     2.289
## SubjID         (Intercept)  4.575     2.139
## Residual                        3.304     1.818
## Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  2.618e-01  2.842e-01  5.700e+01  0.921  0.3608
## family_size1  4.522e-01  6.392e-02  1.618e+03  7.074 2.23e-12 ***
## LP           1.231e-03  2.311e-01  5.700e+01  0.005  0.9958
## OS           6.771e-01  3.179e-01  5.700e+01  2.130  0.0375 *
## family_size1:LP -7.114e-02  5.201e-02  1.618e+03 -1.368  0.1716
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```



```
## Correlation of Fixed Effects:
##          (Intr) fmly_1 LP      OS
## family_siz1  0.000
## LP           0.038  0.000
## OS           -0.046  0.000  0.021
## fmly_siz1:LP  0.000  0.039  0.000  0.000

# Effect sizes for individual terms
eta_squared(m_w_N400_fs, partial = TRUE)

## # Effect Size for ANOVA (Type III)
##
## Parameter      | Eta2 (partial) |      95% CI
## -----
## family_size    |          0.03 | [0.02, 1.00]
## LP             |       4.98e-07 | [0.00, 1.00]
## OS             |          0.07 | [0.00, 1.00]
## family_size:LP |       1.15e-03 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].

# Model R²
r2_nakagawa(m_w_N400_fs)

## # R2 for Mixed Models
##
##      Conditional R2: 0.756
##      Marginal R2: 0.031
```

NON-WORDS

RESPONSE TIME DATA

```
contrasts(rt_nw$complexity) <- c(-0.5, 0.5)

m_rt_nw <- lmer(
  logRT ~ complexity + zLogBF + zLogFS + OS + LP +
    (1 + complexity | SubjID) +
    (1 | ItemID),
  data = rt_nw,
  control = lmerControl(optimizer = "bobyqa")
)
m_rt_nw

## Linear mixed model fit by REML ['lmerModLmerTest']
## Formula: logRT ~ complexity + zLogBF + zLogFS + OS + LP + (1 + complexity |
##      SubjID) + (1 | ItemID)
##      Data: rt_nw
## REML criterion at convergence: -4165.782
## Random effects:
##   Groups   Name                Std.Dev. Corr
##   ItemID   (Intercept) 0.03662
##   SubjID   (Intercept) 0.13143
##           complexity1 0.02365 -0.17
```

```
## Residual 0.14641
## Number of obs: 4607, groups: ItemID, 100; SubjID, 67
## Fixed Effects:
## (Intercept) complexity1 zLogBF zLogFS OS LP
## 6.554486 0.048807 0.015469 -0.006359 -0.041063 0.003398
```

```
summary(m_rt_nw)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: logRT ~ complexity + zLogBF + zLogFS + OS + LP + (1 + complexity |
## SubjID) + (1 | ItemID)
## Data: rt_nw
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: -4165.8
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -4.2605 -0.6625 -0.0227 0.6633 3.8775
##
## Random effects:
## Groups Name Variance Std.Dev. Corr
## ItemID (Intercept) 0.0013408 0.03662
## SubjID (Intercept) 0.0172747 0.13143
## complexity1 0.0005594 0.02365 -0.17
## Residual 0.0214360 0.14641
## Number of obs: 4607, groups: ItemID, 100; SubjID, 67
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 6.554486 0.016639 68.832542 393.923 < 2e-16 ***
## complexity1 0.048807 0.005347 62.675277 9.128 4.07e-13 ***
## zLogBF 0.015469 0.005080 98.214326 3.045 0.00299 **
## zLogFS -0.006359 0.005137 96.636213 -1.238 0.21880
## OS -0.041063 0.018454 62.212150 -2.225 0.02970 *
## LP 0.003398 0.012228 62.906715 0.278 0.78204
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) cmplx1 zLogBF zLogFS OS
## complexity1 -0.073
## zLogBF -0.004 -0.005
## zLogFS -0.004 0.011 -0.576
## OS 0.011 -0.007 -0.001 -0.001
## LP 0.003 -0.015 -0.002 0.000 0.009
```

```
# Effect sizes for individual terms
eta_squared(m_rt_nw, partial = TRUE)
```

```
## # Effect Size for ANOVA (Type III)
##
## Parameter | Eta2 (partial) | 95% CI
## -----
```

```
## complexity |          0.57 | [0.44, 1.00]
## zLogBF      |          0.09 | [0.02, 1.00]
## zLogFS      |          0.02 | [0.00, 1.00]
## OS          |          0.07 | [0.00, 1.00]
## LP          |      1.23e-03 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

```
# Model R2
r2_nakagawa(m_rt_nw)
```

```
## # R2 for Mixed Models
##
##   Conditional R2: 0.494
##   Marginal R2: 0.050
```

ERP DATA

N250 DATA

```
contrasts(erp_bf_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_bf_nw$base_frequency) <- c(-0.5, 0.5)

m_nw_N250_bf <- lmer(
  value ~ complexity * OS + complexity * LP + base_frequency +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_nw, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N250_bf)
```

Base Frequency

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ complexity * OS + complexity * LP + base_frequency +
##   (1 | SubjID) + (1 | SubjID:chlabel)
##   Data: subset(erp_bf_nw, time_window == "N250")
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 41527.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -12.5220  -0.4882  -0.0036   0.4647  24.5889
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
##   SubjID:chlabel (Intercept)  10.462     3.235
##   SubjID          (Intercept)   9.777     3.127
##   Residual                        27.474     5.242
## Number of obs: 6480, groups:  SubjID:chlabel, 1620; SubjID, 60
##
## Fixed effects:
```

```
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -1.4173     0.4175   57.0000   -3.395  0.00126 **
## complexity1    -0.2243     0.1305  4856.0002   -1.719  0.08566 .
## OS              0.5743     0.4669   57.0000    1.230  0.22379
## LP             -0.3205     0.3394   57.0000   -0.944  0.34909
## base_frequency1 -0.1639     0.1302  4856.0002   -1.258  0.20835
## complexity1:OS   0.6090     0.1459  4856.0002    4.173 3.05e-05 ***
## complexity1:LP   0.7230     0.1061  4856.0002    6.816 1.05e-11 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Correlation of Fixed Effects:
```

```
##           (Intr) cmplx1 OS      LP      bs_fr1 cm1:OS
## complexity1  0.000
## OS          -0.046  0.000
## LP           0.038  0.000  0.021
## bas_frncyc1  0.000  0.000  0.000  0.000
## cmplxty1:OS  0.000 -0.046  0.000  0.000  0.000
## cmplxty1:LP  0.000  0.038  0.000  0.000  0.000  0.021
```

```
# Effect sizes for individual terms
```

```
eta_squared(m_nw_N250_bf, partial = TRUE)
```

```
## # Effect Size for ANOVA (Type III)
```

```
##
```

```
## Parameter      | Eta2 (partial) |      95% CI
## -----
## complexity      |      6.08e-04 | [0.00, 1.00]
## OS              |      0.03     | [0.00, 1.00]
## LP              |      0.02     | [0.00, 1.00]
## base_frequency  |      3.26e-04 | [0.00, 1.00]
## complexity:OS   |      3.57e-03 | [0.00, 1.00]
## complexity:LP   |      9.48e-03 | [0.01, 1.00]
```

```
##
```

```
## - One-sided CIs: upper bound fixed at [1.00].
```

```
# Model R²
```

```
r2_nakagawa(m_nw_N250_bf)
```

```
## # R2 for Mixed Models
```

```
##
```

```
## Conditional R2: 0.433
```

```
## Marginal R2: 0.015
```

```
contrasts(erp_fs_nw$complexity) <- c(-0.5, 0.5)
```

```
contrasts(erp_fs_nw$family_size) <- c(-0.5, 0.5)
```

```
m_nw_N250_fs <- lmer(
  value ~ complexity * OS + complexity * LP + family_size +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_nw, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
```

```
summary(m_nw_N250_fs)
```

Family Size

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ complexity * OS + complexity * LP + family_size + (1 |
##   SubjID) + (1 | SubjID:chlabel)
## Data: subset(erp_fs_nw, time_window == "N250")
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 29933.2
##
## Scaled residuals:
##   Min       1Q   Median       3Q      Max
## -7.7366 -0.5107 -0.0056  0.4938 12.2676
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
##   SubjID:chlabel (Intercept)  3.109      1.763
##   SubjID          (Intercept)  2.163      1.471
##   Residual                        4.072      2.018
## Number of obs: 6480, groups:  SubjID:chlabel, 1620; SubjID, 60
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.66509    0.19682   56.99996  -3.379 0.001317 **
## complexity1   -0.23263    0.05023  4856.00002  -4.631 3.73e-06 ***
## OS              0.29808    0.22013   56.99996   1.354 0.181051
## LP             -0.20675    0.16003   56.99996  -1.292 0.201577
## family_size1    0.18172    0.05014  4856.00002   3.625 0.000292 ***
## complexity1:OS  0.39199    0.05618  4856.00002   6.978 3.40e-12 ***
## complexity1:LP  0.35764    0.04084  4856.00002   8.758 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cmplx1 OS      LP      fmly_1 cm1:OS
## complexity1  0.000
## OS          -0.046 0.000
## LP           0.038 0.000 0.021
## family_siz1 0.000 0.000 0.000 0.000
## cmplxty1:OS 0.000 -0.046 0.000 0.000 0.000
## cmplxty1:LP 0.000 0.038 0.000 0.000 0.000 0.021
##
## Effect sizes for individual terms
eta_squared(m_nw_N250_fs, partial = TRUE)
```

```
## # Effect Size for ANOVA (Type III)
##
## Parameter      | Eta2 (partial) |      95% CI
## -----
## complexity      |      4.40e-03 | [0.00, 1.00]
## OS              |      0.03 | [0.00, 1.00]
## LP              |      0.03 | [0.00, 1.00]
```

```
## family_size | 2.70e-03 | [0.00, 1.00]
## complexity:OS | 9.93e-03 | [0.01, 1.00]
## complexity:LP | 0.02 | [0.01, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

```
# Model R2
r2_nakagawa(m_nw_N250_fs)
```

```
## # R2 for Mixed Models
##
## Conditional R2: 0.575
## Marginal R2: 0.025
```

N400 DATA

```
contrasts(erp_bf_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_bf_nw$base_frequency) <- c(-0.5, 0.5)

m_nw_N400_bf <- lmer(
  value ~ complexity * base_frequency * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_nw, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N400_bf)
```

Base Frequency

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ complexity * base_frequency * LP + OS + (1 | SubjID) +
## (1 | SubjID:chlabel)
## Data: subset(erp_bf_nw, time_window == "N400")
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 44167.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -11.9475  -0.3895   0.0032   0.4332  20.9414
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## SubjID:chlabel (Intercept) 23.71     4.869
## SubjID          (Intercept) 23.96     4.895
## Residual                38.00     6.164
## Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
##
## Fixed effects:
##
##              Estimate Std. Error    df t value
## (Intercept)   -0.84542    0.64909 56.99987  -1.302
## complexity1   -0.44211    0.15327 4854.00011  -2.884
## base_frequency1 -0.04368    0.15327 4854.00011  -0.285
```

```
## LP 0.23770 0.52776 56.99987 0.450
## OS 0.74117 0.72597 56.99987 1.021
## complexity1:base_frequency1 -0.05967 0.30655 4854.00011 -0.195
## complexity1:LP 0.20562 0.12473 4854.00011 1.649
## base_frequency1:LP 0.11261 0.12473 4854.00011 0.903
## complexity1:base_frequency1:LP -0.17117 0.24945 4854.00011 -0.686
## Pr(>|t|)
## (Intercept) 0.19799
## complexity1 0.00394 **
## base_frequency1 0.77566
## LP 0.65413
## OS 0.31159
## complexity1:base_frequency1 0.84566
## complexity1:LP 0.09930 .
## base_frequency1:LP 0.36663
## complexity1:base_frequency1:LP 0.49264
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) cmplx1 bs_fr1 LP OS cm1:_1 cm1:LP b_1:LP
## complexity1 0.000
## bas_frqn1 0.000 0.000
## LP 0.038 0.000 0.000
## OS -0.046 0.000 0.000 0.021
## cmplxty1:_1 0.000 0.000 0.000 0.000 0.000
## cmplxty1:LP 0.000 0.039 0.000 0.000 0.000 0.000
## bs_frqn1:LP 0.000 0.000 0.039 0.000 0.000 0.000 0.000
## cmpl1:_1:LP 0.000 0.000 0.000 0.000 0.000 0.039 0.000 0.000
# Effect sizes for individual terms
eta_squared(m_nw_N400_bf, partial = TRUE)
```

```
## # Effect Size for ANOVA (Type III)
##
## Parameter | Eta2 (partial) | 95% CI
## -----|-----|-----
## complexity | 1.71e-03 | [0.00, 1.00]
## base_frequency | 1.67e-05 | [0.00, 1.00]
## LP | 3.55e-03 | [0.00, 1.00]
## OS | 0.02 | [0.00, 1.00]
## complexity:base_frequency | 7.81e-06 | [0.00, 1.00]
## complexity:LP | 5.60e-04 | [0.00, 1.00]
## base_frequency:LP | 1.68e-04 | [0.00, 1.00]
## complexity:base_frequency:LP | 9.70e-05 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

```
# Model R²
r2_nakagawa(m_nw_N400_bf)
```

```
## # R2 for Mixed Models
##
## Conditional R2: 0.559
## Marginal R2: 0.007
```

```

contrasts(erp_fs_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_fs_nw$family_size) <- c(-0.5, 0.5)

m_nw_N400_fs <- lmer(
  value ~ complexity * family_size * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_nw, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N400_fs)

```

Family Size

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ complexity * family_size * LP + OS + (1 | SubjID) + (1 |
## SubjID:chlabel)
## Data: subset(erp_fs_nw, time_window == "N400")
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 32979.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -13.9873  -0.4837  -0.0118   0.4655  14.1595
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## SubjID:chlabel (Intercept) 6.389     2.528
## SubjID         (Intercept) 5.568     2.360
## Residual                        6.114     2.473
## Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.45163    0.31311   57.00004  -1.442 0.154665
## complexity1   -0.45956    0.06148  4854.00021  -7.475 9.11e-14
## family_size1    0.08743    0.06148  4854.00021   1.422 0.155086
## LP              0.01139    0.25458   57.00004   0.045 0.964463
## OS              0.38123    0.35019   57.00004   1.089 0.280904
## complexity1:family_size1 -0.44008    0.12296  4854.00021  -3.579 0.000348
## complexity1:LP    0.25043    0.05003  4854.00021   5.006 5.77e-07
## family_size1:LP    0.12120    0.05003  4854.00021   2.423 0.015450
## complexity1:family_size1:LP 0.22717    0.10006  4854.00021   2.270 0.023230
##
## (Intercept)
## complexity1          ***
## family_size1
## LP
## OS
## complexity1:family_size1 ***
## complexity1:LP      ***

```



```
## family_size1:LP *
## complexity1:family_size1:LP *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) cmplx1 fmly_1 LP      OS      cm1:_1 cm1:LP f_1:LP
## complexity1  0.000
## family_siz1  0.000  0.000
## LP           0.038  0.000  0.000
## OS           -0.046  0.000  0.000  0.021
## cmplxty1:_1  0.000  0.000  0.000  0.000  0.000
## cmplxty1:LP  0.000  0.039  0.000  0.000  0.000  0.000
## fmly_sz1:LP  0.000  0.000  0.039  0.000  0.000  0.000  0.000
## cmpl1:_1:LP  0.000  0.000  0.000  0.000  0.000  0.039  0.000  0.000

# Effect sizes for individual terms
eta_squared(m_nw_N400_fs, partial = TRUE)
```

```
## # Effect Size for ANOVA (Type III)
##
## Parameter | Eta2 (partial) | 95% CI
## -----
## complexity | 0.01 | [0.01, 1.00]
## family_size | 4.16e-04 | [0.00, 1.00]
## LP | 3.51e-05 | [0.00, 1.00]
## OS | 0.02 | [0.00, 1.00]
## complexity:family_size | 2.63e-03 | [0.00, 1.00]
## complexity:LP | 5.14e-03 | [0.00, 1.00]
## family_size:LP | 1.21e-03 | [0.00, 1.00]
## complexity:family_size:LP | 1.06e-03 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

```
# Model R2
r2_nakagawa(m_nw_N400_fs)
```

```
## # R2 for Mixed Models
##
## Conditional R2: 0.666
## Marginal R2: 0.012
```

FOLLOW UP TESTS WORDS

N250

Base Frequency

```
# N250 Words: Base Frequency x OS
emm_bf_OS <- emmeans(m_w_N250_bf,
  ~ base_frequency | OS,
  at = list(OS = c(-1, 0, 1)))
```

```
## Note: D.f. calculations have been disabled because the number of observations exceeds 3000.
## To enable adjustments, add the argument 'pbkrtest.limit = 3240' (or larger)
## [or, globally, 'set emm_options(pbkrtest.limit = 3240)' or larger];
```

```
## but be warned that this may result in large computation time and memory use.
## Note: D.f. calculations have been disabled because the number of observations exceeds 3000.
## To enable adjustments, add the argument 'lmerTest.limit = 3240' (or larger)
## [or, globally, 'set emm_options(lmerTest.limit = 3240)' or larger];
## but be warned that this may result in large computation time and memory use.
```

```
summary(emm_bf_OS)
```

```
## OS = -1:
## base_frequency emmean SE df asymp.LCL asymp.UCL
## Low -2.082 0.718 Inf -3.49 -0.67611
## High -1.946 0.718 Inf -3.35 -0.53951
##
## OS = 0:
## base_frequency emmean SE df asymp.LCL asymp.UCL
## Low -1.321 0.467 Inf -2.24 -0.40492
## High -1.639 0.467 Inf -2.56 -0.72327
##
## OS = 1:
## base_frequency emmean SE df asymp.LCL asymp.UCL
## Low -0.559 0.685 Inf -1.90 0.78312
## High -1.333 0.685 Inf -2.67 0.00982
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
pairs(emm_bf_OS)
```

```
## OS = -1:
## contrast estimate SE df z.ratio p.value
## Low - High -0.137 0.154 Inf -0.889 0.3740
##
## OS = 0:
## contrast estimate SE df z.ratio p.value
## Low - High 0.318 0.100 Inf 3.181 0.0015
##
## OS = 1:
## contrast estimate SE df z.ratio p.value
## Low - High 0.773 0.147 Inf 5.273 <.0001
##
## Degrees-of-freedom method: asymptotic
```

N400

```
# N400 Words: Base Frequency x LP
emm_bf_LP <- emmeans(m_w_N400_bf,
  ~ base_frequency | LP,
  at = list(LP = c(-1, 0, 1)))
```

Base Frequency

```
## Note: D.f. calculations have been disabled because the number of observations exceeds 3000.
## To enable adjustments, add the argument 'pbkrtest.limit = 3240' (or larger)
## [or, globally, 'set emm_options(pbkrtest.limit = 3240)' or larger];
## but be warned that this may result in large computation time and memory use.
```

```
## Note: D.f. calculations have been disabled because the number of observations exceeds 3000.
## To enable adjustments, add the argument 'lmerTest.limit = 3240' (or larger)
## [or, globally, 'set emm_options(lmerTest.limit = 3240)' or larger];
## but be warned that this may result in large computation time and memory use.
```

```
summary(emm_bf_LP)
```

```
## LP = -1:
## base_frequency emmean    SE df asymp.LCL asymp.UCL
## Low           0.634 0.726 Inf   -0.788    2.06
## High          0.504 0.726 Inf   -0.919    1.93
##
## LP = 0:
## base_frequency emmean    SE df asymp.LCL asymp.UCL
## Low           0.509 0.574 Inf   -0.616    1.63
## High          0.723 0.574 Inf   -0.402    1.85
##
## LP = 1:
## base_frequency emmean    SE df asymp.LCL asymp.UCL
## Low           0.383 0.754 Inf   -1.095    1.86
## High          0.942 0.754 Inf   -0.537    2.42
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
pairs(emm_bf_LP)
```

```
## LP = -1:
## contrast estimate    SE df z.ratio p.value
## Low - High    0.130 0.135 Inf    0.968  0.3328
##
## LP = 0:
## contrast estimate    SE df z.ratio p.value
## Low - High   -0.214 0.106 Inf   -2.012  0.0442
##
## LP = 1:
## contrast estimate    SE df z.ratio p.value
## Low - High   -0.558 0.140 Inf   -3.995  0.0001
##
## Degrees-of-freedom method: asymptotic
```

FOLLOW UP TESTS NON-WORDS

ERP DATA

N250

```
# Complexity effect at three levels of OS
emm_complexity_OS_bf <- emmeans(m_nw_N250_bf,
                                ~ complexity | OS,
                                at = list(OS = c(-1, 0, 1)),pbkrtest.limit = 6480)
summary(emm_complexity_OS_bf)
```

Base Frequency

```
## OS = -1:
## complexity emmean SE df lower.CL upper.CL
## Simple -1.542 0.648 59.8 -2.84 -0.245
## Complex -2.410 0.648 59.8 -3.71 -1.113
##
## OS = 0:
## complexity emmean SE df lower.CL upper.CL
## Simple -1.272 0.422 59.8 -2.12 -0.428
## Complex -1.531 0.422 59.8 -2.38 -0.687
##
## OS = 1:
## complexity emmean SE df lower.CL upper.CL
## Simple -1.002 0.619 59.8 -2.24 0.235
## Complex -0.653 0.619 59.8 -1.89 0.585
##
## Results are averaged over the levels of: base_frequency
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
# Pairwise complexity contrast at each level of OS
pairs(emm_complexity_OS_bf)
```

```
## OS = -1:
## contrast estimate SE df t.ratio p.value
## Simple - Complex 0.868 0.200 4856 4.337 <.0001
##
## OS = 0:
## contrast estimate SE df t.ratio p.value
## Simple - Complex 0.259 0.130 4856 1.988 0.0469
##
## OS = 1:
## contrast estimate SE df t.ratio p.value
## Simple - Complex -0.350 0.191 4856 -1.831 0.0672
##
## Results are averaged over the levels of: base_frequency
## Degrees-of-freedom method: kenward-roger
```

```
# Complexity effect at three levels of LP
emm_complexity_LP_bf <- emmeans(m_nw_N250_bf,
                                ~ complexity | LP,
                                at = list(LP = c(-1, 0, 1)),pbkrtest.limit = 6480)
summary(emm_complexity_LP_bf)
```

```
## LP = -1:
## complexity emmean SE df lower.CL upper.CL
## Simple -0.612 0.534 59.8 -1.68 0.456
## Complex -1.534 0.534 59.8 -2.60 -0.466
##
## LP = 0:
## complexity emmean SE df lower.CL upper.CL
## Simple -1.294 0.422 59.8 -2.14 -0.449
## Complex -1.493 0.422 59.8 -2.34 -0.648
##
## LP = 1:
## complexity emmean SE df lower.CL upper.CL
## Simple -1.976 0.555 59.8 -3.09 -0.866
```

```
## Complex    -1.452 0.555 59.8    -2.56    -0.342
##
## Results are averaged over the levels of: base_frequency
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
# Pairwise complexity contrast at each level of LP
pairs(emm_complexity_LP_bf)

## LP = -1:
## contrast      estimate      SE    df t.ratio p.value
## Simple - Complex    0.922 0.165 4856    5.594 <.0001
##
## LP = 0:
## contrast      estimate      SE    df t.ratio p.value
## Simple - Complex    0.199 0.130 4856    1.526 0.1272
##
## LP = 1:
## contrast      estimate      SE    df t.ratio p.value
## Simple - Complex   -0.524 0.171 4856   -3.061 0.0022
##
## Results are averaged over the levels of: base_frequency
## Degrees-of-freedom method: kenward-roger
```

Family Size

```
# Complexity effect at three levels of OS
emm_complexity_OS_fs <- emmeans(m_nw_N250_fs,
                                ~ complexity | OS,
                                at = list(OS = c(-1, 0, 1)),pbkrtest.limit = 6480)
summary(emm_complexity_OS_fs)
```

Complexity x OS interaction

```
## OS = -1:
## complexity emmean      SE    df lower.CL upper.CL
## Simple     -0.632 0.304 58.9    -1.241   -0.0231
## Complex    -1.274 0.304 58.9    -1.883   -0.6650
##
## OS = 0:
## complexity emmean      SE    df lower.CL upper.CL
## Simple     -0.530 0.198 58.9    -0.927   -0.1334
## Complex    -0.780 0.198 58.9    -1.177   -0.3833
##
## OS = 1:
## complexity emmean      SE    df lower.CL upper.CL
## Simple     -0.428 0.291 58.9    -1.010    0.1534
## Complex    -0.286 0.291 58.9    -0.867    0.2955
##
## Results are averaged over the levels of: family_size
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
# Pairwise complexity contrast at each level of OS
pairs(emm_complexity_OS_fs)
```

```
## OS = -1:
## contrast      estimate      SE    df t.ratio p.value
## Simple - Complex    0.642 0.0771 4856   8.329 <.0001
##
## OS = 0:
## contrast      estimate      SE    df t.ratio p.value
## Simple - Complex    0.250 0.0502 4856   4.979 <.0001
##
## OS = 1:
## contrast      estimate      SE    df t.ratio p.value
## Simple - Complex   -0.142 0.0736 4856  -1.932  0.0534
##
## Results are averaged over the levels of: family_size
## Degrees-of-freedom method: kenward-roger
```

Complexity × LP Interaction

```
# Complexity effect at three levels of LP
emm_complexity_LP_fs <- emmeans(m_nw_N250_fs,
                                ~ complexity | LP,
                                at = list(LP = c(-1, 0, 1)), pbkrtest.limit = 6480)
summary(emm_complexity_LP_fs)
```

```
## LP = -1:
## complexity emmean      SE    df lower.CL upper.CL
## Simple     -0.159 0.251 58.9   -0.660    0.343
## Complex    -0.733 0.251 58.9   -1.234   -0.231
##
## LP = 0:
## complexity emmean      SE    df lower.CL upper.CL
## Simple     -0.545 0.198 58.9   -0.941   -0.148
## Complex    -0.761 0.198 58.9   -1.157   -0.364
##
## LP = 1:
## complexity emmean      SE    df lower.CL upper.CL
## Simple     -0.930 0.260 58.9   -1.451   -0.409
## Complex    -0.789 0.260 58.9   -1.310   -0.268
##
## Results are averaged over the levels of: family_size
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
# Pairwise complexity contrast at each level of LP
pairs(emm_complexity_LP_fs)
```

```
## LP = -1:
## contrast      estimate      SE    df t.ratio p.value
## Simple - Complex    0.574 0.0634 4856   9.047 <.0001
##
## LP = 0:
## contrast      estimate      SE    df t.ratio p.value
## Simple - Complex    0.216 0.0502 4856   4.310 <.0001
##
## LP = 1:
## contrast      estimate      SE    df t.ratio p.value
```

```
## Simple - Complex    -0.141 0.0659 4856   -2.145   0.0320
##
## Results are averaged over the levels of: family_size
## Degrees-of-freedom method: kenward-roger
```

N400

Family Size

```
# Step 1: Unpack the three-way
# Complexity x Family Size at three levels of LP
emm_3way <- emmeans(m_nw_N400_fs,
                    ~ complexity * family_size | LP,
                    at = list(LP = c(-1, 0, 1)), pbkrtest.limit = 6480)
summary(emm_3way)
```

Complexity × Family Size × LP interaction

```
## LP = -1:
## complexity family_size emmean SE df lower.CL upper.CL
## Simple Small -0.2420 0.401 60.4 -1.044 0.5603
## Complex Small -0.6184 0.401 60.4 -1.421 0.1839
## Simple Large 0.0578 0.401 60.4 -0.744 0.8602
## Complex Large -0.9858 0.401 60.4 -1.788 -0.1835
##
## LP = 0:
## complexity family_size emmean SE df lower.CL upper.CL
## Simple Small -0.3596 0.317 60.4 -0.994 0.2749
## Complex Small -0.5992 0.317 60.4 -1.234 0.0354
## Simple Large -0.0522 0.317 60.4 -0.687 0.5824
## Complex Large -0.7318 0.317 60.4 -1.366 -0.0972
##
## LP = 1:
## complexity family_size emmean SE df lower.CL upper.CL
## Simple Small -0.4773 0.417 60.4 -1.311 0.3565
## Complex Small -0.5799 0.417 60.4 -1.414 0.2538
## Simple Large -0.1622 0.417 60.4 -0.996 0.6716
## Complex Large -0.4778 0.417 60.4 -1.312 0.3560
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
# Step 2: Pairwise complexity contrasts within each
# family size condition at each level of LP
pairs(emm_3way, simple = "complexity")
```

```
## family_size = Small, LP = -1:
## contrast estimate SE df t.ratio p.value
## Simple - Complex 0.376 0.1100 4854 3.424 0.0006
##
## family_size = Large, LP = -1:
## contrast estimate SE df t.ratio p.value
## Simple - Complex 1.044 0.1100 4854 9.494 <.0001
##
## family_size = Small, LP = 0:
```

```

## contrast      estimate      SE    df t.ratio p.value
## Simple - Complex    0.240 0.0869 4854    2.755  0.0059
##
## family_size = Large, LP = 0:
## contrast      estimate      SE    df t.ratio p.value
## Simple - Complex    0.680 0.0869 4854    7.816  <.0001
##
## family_size = Small, LP = 1:
## contrast      estimate      SE    df t.ratio p.value
## Simple - Complex    0.103 0.1140 4854    0.899  0.3688
##
## family_size = Large, LP = 1:
## contrast      estimate      SE    df t.ratio p.value
## Simple - Complex    0.316 0.1140 4854    2.763  0.0058
##
## Degrees-of-freedom method: kenward-roger

# Step 3: Pairwise family size contrasts within each
# complexity condition at each level of LP
pairs(emm_3way, simple = "family_size")

## complexity = Simple, LP = -1:
## contrast      estimate      SE    df t.ratio p.value
## Small - Large   -0.300 0.1100 4854   -2.728  0.0064
##
## complexity = Complex, LP = -1:
## contrast      estimate      SE    df t.ratio p.value
## Small - Large    0.367 0.1100 4854    3.342  0.0008
##
## complexity = Simple, LP = 0:
## contrast      estimate      SE    df t.ratio p.value
## Small - Large   -0.307 0.0869 4854   -3.536  0.0004
##
## complexity = Complex, LP = 0:
## contrast      estimate      SE    df t.ratio p.value
## Small - Large    0.133 0.0869 4854    1.525  0.1273
##
## complexity = Simple, LP = 1:
## contrast      estimate      SE    df t.ratio p.value
## Small - Large   -0.315 0.1140 4854   -2.758  0.0058
##
## complexity = Complex, LP = 1:
## contrast      estimate      SE    df t.ratio p.value
## Small - Large   -0.102 0.1140 4854   -0.894  0.3712
##
## Degrees-of-freedom method: kenward-roger

# Step 4: The interaction contrast - does the complexity x
# family size interaction itself change across LP levels?
# This is the formal test of the three-way
emm_2way_by_LP <- emmeans(m_nw_N400_fs,
  ~ complexity * family_size | LP,
  at = list(LP = c(-1, 0, 1)), pbkrtest.limit = 6480)
contrast(emm_2way_by_LP,
  interaction = c("pairwise", "pairwise"))

```



```
## LP = -1:
## complexity_pairwise family_size_pairwise estimate SE df t.ratio p.value
## Simple - Complex Small - Large -0.667 0.155 4854 -4.292 <.0001
##
## LP = 0:
## complexity_pairwise family_size_pairwise estimate SE df t.ratio p.value
## Simple - Complex Small - Large -0.440 0.123 4854 -3.579 0.0003
##
## LP = 1:
## complexity_pairwise family_size_pairwise estimate SE df t.ratio p.value
## Simple - Complex Small - Large -0.213 0.162 4854 -1.318 0.1876
##
## Degrees-of-freedom method: kenward-roger
```

```
# Complexity x Family Size averaged over LP
emm_2way <- emmeans(m_nw_N400_fs, ~ complexity * family_size, pbkrtest.limit = 6480)
```

Complexity × Family Size interaction averaged across LP

NOTE: Results may be misleading due to involvement in interactions

```
summary(emm_2way)
```

```
## complexity family_size emmean SE df lower.CL upper.CL
## Simple Small -0.3540 0.317 60.4 -0.988 0.280
## Complex Small -0.6001 0.317 60.4 -1.234 0.034
## Simple Large -0.0469 0.317 60.4 -0.681 0.587
## Complex Large -0.7440 0.317 60.4 -1.378 -0.110
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
pairs(emm_2way, simple = "complexity")
```

```
## family_size = Small:
## contrast estimate SE df t.ratio p.value
## Simple - Complex 0.246 0.0869 4854 2.833 0.0046
##
## family_size = Large:
## contrast estimate SE df t.ratio p.value
## Simple - Complex 0.697 0.0869 4854 8.025 <.0001
##
## Degrees-of-freedom method: kenward-roger
```

```
pairs(emm_2way, simple = "family_size")
```

```
## complexity = Simple:
## contrast estimate SE df t.ratio p.value
## Small - Large -0.307 0.0869 4854 -3.535 0.0004
##
## complexity = Complex:
## contrast estimate SE df t.ratio p.value
## Small - Large 0.144 0.0869 4854 1.657 0.0976
##
## Degrees-of-freedom method: kenward-roger
```

Complexity × LP interaction averaged over Family Size:

```
# Complexity x LP averaged over FS at three levels of LP
emm_complexity_LP_N400 <- emmeans(m_nw_N400_fs,
                                   ~ complexity | LP,
                                   at = list(LP = c(-1, 0, 1)), pbkrtest.limit = 6480)
```

NOTE: Results may be misleading due to involvement in interactions

```
summary(emm_complexity_LP_N400)
```

```
## LP = -1:
## complexity emmean SE df lower.CL upper.CL
## Simple -0.0921 0.397 58.1 -0.887 0.7033
## Complex -0.8021 0.397 58.1 -1.597 -0.0067
##
## LP = 0:
## complexity emmean SE df lower.CL upper.CL
## Simple -0.2059 0.314 58.1 -0.835 0.4232
## Complex -0.6655 0.314 58.1 -1.295 -0.0364
##
## LP = 1:
## complexity emmean SE df lower.CL upper.CL
## Simple -0.3197 0.413 58.1 -1.146 0.5068
## Complex -0.5289 0.413 58.1 -1.355 0.2977
##
## Results are averaged over the levels of: family_size
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
pairs(emm_complexity_LP_N400)
```

```
## LP = -1:
## contrast estimate SE df t.ratio p.value
## Simple - Complex 0.710 0.0777 4854 9.135 <.0001
##
## LP = 0:
## contrast estimate SE df t.ratio p.value
## Simple - Complex 0.460 0.0615 4854 7.475 <.0001
##
## LP = 1:
## contrast estimate SE df t.ratio p.value
## Simple - Complex 0.209 0.0808 4854 2.589 0.0096
##
## Results are averaged over the levels of: family_size
## Degrees-of-freedom method: kenward-roger
```

FS × LP interaction averaged over Complexity:

```
# Complexity x LP averaged over FS at three levels of LP
emm_FS_LP_N400 <- emmeans(m_nw_N400_fs,
                           ~ family_size | LP,
                           at = list(LP = c(-1, 0, 1)), pbkrtest.limit = 6480)
```

NOTE: Results may be misleading due to involvement in interactions

```
summary(emm_FS_LP_N400)
```

```
## LP = -1:
## family_size emmean    SE    df lower.CL upper.CL
## Small      -0.430 0.397 58.1   -1.23    0.365
## Large      -0.464 0.397 58.1   -1.26    0.331
##
## LP = 0:
## family_size emmean    SE    df lower.CL upper.CL
## Small      -0.479 0.314 58.1   -1.11    0.150
## Large      -0.392 0.314 58.1   -1.02    0.237
##
## LP = 1:
## family_size emmean    SE    df lower.CL upper.CL
## Small      -0.529 0.413 58.1   -1.36    0.298
## Large      -0.320 0.413 58.1   -1.15    0.507
##
## Results are averaged over the levels of: complexity
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
pairs(emm_FS_LP_N400)
```

```
## LP = -1:
## contrast      estimate    SE    df t.ratio p.value
## Small - Large    0.0338 0.0777 4854    0.434  0.6640
##
## LP = 0:
## contrast      estimate    SE    df t.ratio p.value
## Small - Large  -0.0874 0.0615 4854   -1.422  0.1551
##
## LP = 1:
## contrast      estimate    SE    df t.ratio p.value
## Small - Large  -0.2086 0.0808 4854   -2.583  0.0098
##
## Results are averaged over the levels of: complexity
## Degrees-of-freedom method: kenward-roger
```